

STUDY GUIDE

Male Reproductive Physiology

HPG Axis · Testosterone · Spermatogenesis · Erection · Hypogonadism

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PHS 220 — Physiology

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§1 ANATOMY OF THE MALE REPRODUCTIVE SYSTEM

Structure	Key Anatomy	Function
Testes (×2)	Oval organs (~4.5 × 3 × 2.5 cm; ~20 mL volume) within scrotum (3–4°C below core — essential for spermatogenesis); 200–300 lobules each containing 2–4 seminiferous tubules	Spermatogenesis; steroidogenesis (testosterone)
Epididymis	~6 m coiled duct lying posterolateral to testis; head (caput), body, tail	Sperm maturation (acquire motility and fertilising capacity); sperm storage (tail)
Vas Deferens	~45 cm muscular duct from epididymis tail → ejaculatory duct; crosses inguinal canal (within spermatic cord)	Transport of sperm; site of vasectomy
Ejaculatory Ducts (×2)	Junction of vas deferens and seminal vesicle duct → pierce prostate → open at verumontanum (prostatic urethra)	Deliver sperm and seminal fluid into urethra
Seminal Vesicles (×2)	Paired coiled glands posterior to bladder	Provide ~60% of semen volume: fructose (sperm energy); prostaglandins; coagulation factors; alkaline pH
Prostate Gland	~20 g; surrounds proximal urethra; 30–50 glands embedded in fibromuscular stroma	~35% of semen volume: PSA (liquefaction); zinc (antibacterial); acid phosphatase; milky fluid (sperm protection)
Bulbourethral (Cowper's) Glands (×2)	Pea-sized; in urogenital diaphragm	Alkaline pre-ejaculatory mucus (neutralises urethral acidity; lubrication)
Scrotum	Pigmented skin + dartos muscle; contains testes	Thermoregulation (cremaster muscle + pampiniform plexus); keeps testes 3–4°C below core temperature

THE SPERMATIC CORD

The spermatic cord contains (from deep to superficial): Vas deferens; Testicular artery (from aorta); Pampiniform plexus of veins (counter-current heat exchange — cools arterial blood entering testis); Lymphatics; Ilioinguinal nerve; Cremasteric muscle; Genital branch of genitofemoral nerve. Varicocele (dilated pampiniform plexus) impairs counter-current cooling → elevated testicular temperature → impaired spermatogenesis — a common cause of male subfertility.

§2 THE HYPOTHALAMO-PITUITARY-GONADAL (HPG) AXIS

Level	Hormone	Stimulus	Key Action	Feedback
Hypothalamus	GnRH (pulsatile, 1 pulse/1-2h)	Kisspeptin from arcuate nucleus; testosterone deficiency	Stimulates LH and FSH from pituitary gonadotrophs	Testosterone (long-loop negative feedback suppresses GnRH)
Anterior Pituitary	LH	GnRH pulses	Stimulates Leydig cells → testosterone synthesis (via cAMP → StAR → cholesterol → testosterone)	Testosterone inhibits LH (negative feedback at pituitary and hypothalamus)
Anterior Pituitary	FSH	GnRH pulses	Stimulates Sertoli cells → support spermatogenesis; ↑ androgen-binding protein (ABP); ↑ inhibin B	Inhibin B (from Sertoli cells) selectively inhibits FSH (not LH) — spermatogenesis-specific feedback
Testis	Testosterone	LH (and constitutive)	Spermatogenesis; secondary sex characteristics; anabolic effects; feedback inhibition of HPG axis	Negative feedback to hypothalamus (↓ GnRH) and pituitary (↓ LH/FSH response)

§3 TESTOSTERONE: PRODUCTION & ACTIONS

3.1 Synthesis in Leydig Cells

Testosterone is a C-19 steroid synthesised from cholesterol in Leydig (interstitial) cells. Leydig cells cluster in the interstitial spaces between seminiferous tubules. LH binding to its receptor (Gs-coupled → ↑ cAMP) stimulates the rate-limiting step: StAR (steroidogenic acute regulatory protein) — transport of cholesterol across the inner mitochondrial membrane.

Conversion	Enzyme	Location
Cholesterol → Pregnenolone	CYP11A1 (side-chain cleavage)	Inner mitochondrial membrane (rate-limiting; StAR)
Pregnenolone → 17-OH-Pregnenolone	CYP17A1 (17 α -hydroxylase)	SER
17-OH-Pregnenolone → DHEA	CYP17A1 (17,20-lyase)	SER
DHEA → Androstenedione	3 β -HSD	SER/Mitochondria
Androstenedione → Testosterone	17 β -HSD	SER

3.2 Transport of Testosterone

Fraction	%	Details
SHBG-bound (tightly)	~60%	Biologically inactive; SHBG produced by liver; ↑ by oestrogen, thyroid hormones; ↓ by androgens, obesity, insulin
Albumin-bound (loosely)	~38%	Weakly bound; dissociates easily → bioavailable
Free testosterone	~2%	Unbound; biologically active; diffuses into cells; measured as "free T"

3.3 Physiological Actions of Testosterone

Phase/System	Action
Embryonic (weeks 8–18)	DHT: external genitalia virilisation (under-5α-reductase); Testosterone: Wolffian duct development → epididymis, vas, seminal vesicles, ejaculatory ducts
Puberty	↑ Testis size; pubic/axillary hair; penile growth; voice deepening; ↑ muscle mass; ↑ linear growth; epiphyseal fusion; libido development; sebaceous gland activity (acne)
Adult male	Maintains spermatogenesis (high intratesticular T essential); libido and sexual function; maintains muscle mass and bone density; erythropoiesis (↑ EPO); sebaceous glands; prostate growth
CNS	Affects behaviour (aggression, libido); converted to E2 in brain (aromatisation) — important for male bone health and negative HPG feedback
Bone	Direct anabolic effect on bone (↑ osteoblasts); indirect via aromatisation to E2 (↑ bone density, epiphyseal fusion)
Erythropoiesis	↑ EPO production; ↑ RBC mass (haematocrit ~45% in men vs ~38% in women) — explains higher haemoglobin in males

TESTOSTERONE vs DHT vs OESTRADIOL IN MALES

Testosterone (T): Principal androgen; circulates and enters cells; acts directly via androgen receptor (AR) in muscle, kidney, prostate.

DHT (Dihydrotestosterone): Formed from T by 5α-reductase (type 1 in skin/liver; type 2 in prostate, external genitalia, hair follicles). 3× more potent than T. Mediates: prostate growth; male pattern baldness; external genitalia virilisation; facial/body hair. **5α-reductase inhibitors (finasteride, dutasteride)** — used for BPH and male pattern baldness.

Oestradiol (E2) from T: Formed by aromatase (CYP19) in fat, liver, brain. Essential for: bone density (male osteoporosis ≈ E2 deficiency); male libido (E2 acts in brain); negative HPG feedback in males.

§4 SPERMATOGENESIS

Spermatogenesis is the process by which diploid spermatogonia (stem cells) are transformed into mature haploid spermatozoa within the seminiferous tubules. The complete process takes approximately 74 days (+ 12–21 days epididymal maturation) and is continuous from puberty onwards.

Stage	Cell	Ploidy	Key Event	Duration
Spermatogonial division	Type A spermatogonia (stem cells) → Type B spermatogonia	Diploid (2n)	Mitotic amplification; self-renewal of stem cell pool; FSH promotes B spermatogonial differentiation	~16 days
Primary spermatocyte	Type B → Primary spermatocyte → enters meiosis I	Diploid (2n, 4C — DNA replicated)	DNA replication; crossing over during prophase I; longest stage	~24 days (in meiosis I)
Secondary spermatocyte	Completion of meiosis I → Secondary spermatocyte	Haploid (n, 2C)	Brief; enters meiosis II immediately	Hours
Spermatid	Meiosis II → Spermatid	Haploid (n, 1C)	Haploid; non-dividing; undergoes spermiogenesis	~24 days
Spermiogenesis	Round spermatid → elongating spermatid → spermatozoon	Haploid	Acrosome formation (from Golgi); flagellum development; nuclear condensation; cytoplasm shedding (residual body phagocytosed by Sertoli cells)	Part of 24-day spermatid phase
Spermiation	Mature spermatozoon released into tubular lumen	Haploid	Androgen-dependent; FSH-supported; immotile at this stage	—

4.1 Sertoli Cell Functions

Sertoli cells are tall columnar 'nurse cells' that span from the basement membrane to the tubular lumen. They do not divide after puberty and each one supports ~48 spermatids. They are essential for spermatogenesis:

Function	Details
Blood-testis barrier	Tight junctions between adjacent Sertoli cells divide tubule into basal (spermatogonia) and adluminal (meiotic/post-meiotic cells) compartments; protects developing sperm from autoimmune attack
Nutritional support	Provide nutrients (lactate, pyruvate) and growth factors to developing germ cells (avascular seminiferous tubules — no blood vessels within)
Phagocytosis	Remove residual bodies (excess cytoplasm) shed during spermiogenesis and apoptotic germ cells
Hormone secretion	Produce inhibin B (↓ FSH); Androgen-Binding Protein (ABP — concentrates testosterone in tubule); Müllerian Inhibiting Substance (MIS/AMH — in foetal life)
FSH response	Primary target of FSH; FSH → ↑ aromatase, ↑ ABP, ↑ inhibin B, ↑ nutrient supply → quantitatively supports spermatogenesis

§5 SPERM BIOLOGY & SEMEN

5.1 Sperm Structure

Region	Structure	Function
Head	~5 µm; elongated nucleus with highly condensed chromatin; covered by acrosomal cap (modified Golgi-derived lysosome containing hydrolytic enzymes)	Contains genetic material; acrosome enzymes (hyaluronidase, acrosin, proteases) digest zona pellucida for fertilisation
Neck	Basal body (9+2 axoneme begins); mitochondrial sheath begins	Connection; energy transduction
Midpiece	~5 µm; tightly coiled mitochondria around axoneme	Energy production (ATP) for flagellar movement
Principal Piece	~45 µm; 9+2 axoneme; fibrous sheath	Flagellar propulsion
End Piece	~5 µm; only axoneme	Terminal segment

5.2 Semen Analysis (WHO Reference Values)

Parameter	WHO Lower Reference Limit (5th centile)
Ejaculate volume	≥1.5 mL
Total sperm number per ejaculate	≥39 million
Sperm concentration	≥16 million/mL
Total motility (PR + NP)	≥42%
Progressive motility (PR)	≥30%
Vitality (live sperm)	≥54%
Normal morphology (Kruger strict)	≥4%
pH	≥7.2

5.3 Capacitation & the Acrosome Reaction

Capacitation is the physiological process by which sperm become capable of fertilisation. It occurs in the female reproductive tract (takes ~7 hours) and involves removal of cholesterol from the sperm plasma membrane, hyperpolarisation then Ca^{2+} influx, and initiation of hyperactivated motility (whip-like flagellar motion).

The acrosome reaction occurs when a capacitated sperm contacts the zona pellucida (ZP3 binds sperm receptor). This triggers fusion of the acrosomal membrane with the plasma membrane → release of acrosomal enzymes (hyaluronidase, acrosin) → penetration of zona pellucida. Only one sperm enters the oocyte; this triggers the cortical reaction (cortical granule

exocytosis → zona hardening) — the block to polyspermy.

§6 ERECTION & EJACULATION

6.1 Physiology of Penile Erection

Penile erection is primarily a vascular event mediated by the parasympathetic nervous system, with psychological, reflex, and nocturnal components.

Step	Event
1. Sexual stimulation	Psychogenic (cortex → spinal cord) or reflexogenic (tactile → sacral reflex arc S2-S4) → parasympathetic outflow via pelvic splanchnic nerves (S2-S4)
2. NO release	Parasympathetic → release of nitric oxide (NO) from endothelium and non-adrenergic non-cholinergic (NANC) nerve terminals; VIP and ACh also play roles
3. cGMP elevation	NO → guanylyl cyclase → ↑ cGMP → activates PKG → smooth muscle relaxation in helicine arteries and sinusoids of corpora cavernosa
4. Vascular engorgement	Relaxed sinusoids fill with blood; expanding corpora compress venous outflow (subtunical venules) → venous occlusive mechanism → rigid erection
5. Detumescence	Sympathetic discharge → NE → α1-AR → smooth muscle contraction → venous drainage restored; PDE5 (phosphodiesterase type 5) degrades cGMP — basis of PDE5 inhibitors (sildenafil, tadalafil) for erectile dysfunction

6.2 Ejaculation

Phase	Neural Control	Event
Emission	Sympathetic (T10-L2); hypogastric nerve	Peristaltic contractions of epididymis, vas deferens, seminal vesicles, prostate → propel semen into prostatic urethra; bladder neck closes (prevents retrograde ejaculation)
Ejaculation (propulsive)	Somatic (S2-S4); pudendal nerve	Rhythmic contractions of bulbocavernosus and ischiocavernosus muscles → semen expelled through urethra
Orgasm	CNS	Sensation of orgasm perceived cortically; timed with ejaculation

§7 MALE HYPOGONADISM

Type	Definition	LH/FSH	Testosterone	Causes & Examples
Primary Hypogonadism (Hypergonadotropic)	Testicular failure → ↓T → ↑ LH/FSH (loss of negative feedback)	HIGH	LOW	Klinefelter syndrome (47,XXY — most common genetic cause); cryptorchidism; orchitis (mumps); chemotherapy/radiotherapy; testicular torsion (bilateral); varicocele

Type	Definition	LH/FSH	Testosterone	Causes & Examples
Secondary (Central) Hypogonadism (Hypogonadotropic)	Hypothalamic or pituitary failure → ↓ GnRH/LH/FSH → ↓ T	LOW or inappropriately normal	LOW	Kallmann syndrome (GnRH deficiency + anosmia); hyperprolactinaemia; panhypopituitarism; anabolic steroid abuse (suppresses HPG axis); opiates; haemochromatosis
Late-Onset Hypogonadism	Age-related decline in testosterone	Mildly ↑	LOW or low-normal	Ageing; obesity (adipose aromatase ↑ E2 → ↓ LH); T2DM

FEATURES OF TESTOSTERONE DEFICIENCY IN ADULT MALES

Sexual: ↓ Libido; erectile dysfunction; reduced ejaculate volume; infertility.

Physical: ↓ Muscle mass and strength; ↑ body fat (especially central); reduced body/pubis/facial hair; fine skin; gynaecomastia (↑ E2:T ratio).

Bone: Osteoporosis (testosterone — via E2 — maintains bone density in males).

Mood/cognition: Fatigue; depression; poor concentration.

Haematological: Mild normocytic anaemia (↓ EPO stimulus).

§8 VASECTOMY & MALE CONTRACEPTION

Vasectomy involves surgical interruption (division and ligation) of the vas deferens bilaterally, preventing sperm transport from the epididymis to the ejaculatory duct. It is a highly effective method of contraception (failure rate <0.1%).

Aspect	Details
Procedure	Under local anaesthesia; small scrotal incision; vas identified, ligated and divided (or occluded by electrocautery, clips); outpatient
Effect on hormones	NO effect on LH, FSH, or testosterone levels — Leydig cell function unaffected; HPG axis intact
Effect on semen	Semen still produced from seminal vesicles and prostate (~98% of ejaculate volume); azoospermia (no sperm in ejaculate)
Contraceptive efficacy	Effective once semen analysis confirms azoospermia (usually after 20 ejaculations or 3 months)
Reversal (vasovasostomy)	Success decreases with time: 90% if <3 years; 50% if >9 years (due to obstructive azoospermia and anti-sperm antibody development)